UNIVERSIDAD TECNICA
FEDERICO SANTA MARIA

Master's Thesis

Design and Sizing of an Energy Storage System for a Hybrid Tugboat

Thesis as a requirement to obtain the degree of
Magister en Ciencias de la Ingeniería Electrónica

Thesis presented by
Leonardo Solis Zamora

Guide Professor
Dr. Marcelo Pérez Leiva

Comisión evaluadora
Dr. Christian Rojas, Correferente, UTFSM
Msc. Joel Perez, Correferente, UACh
Dr. Jon Are Suul, Correferente

Enero 2025 , Valparaíso, Chile

UNIVERSIDAD TECNICA
FEDERICO SANTA MARIA

CONSTANCIA DE VALIDACIÓN Y CONFIDENCIALIDAD DE MONOGRAFÍA A REPOSITORIO ACADÉMICO

1.- IDENTIFICACIÓN DEL TRABAJO ACADÉMICO

Tipo de monografía (marcar una opción): ☐ Memoria o trabajo de título ☐ Tesis de Postgrado
Título del trabajo:
Nombre del candidato(a):
Carrera / Grado:
Campus: Casa Central Valparaíso Departamento: Electrónica

2.- VALIDACIÓN DEL PROFESOR GUÍA/DIRECTOR DE TESIS

Yo, Marcelo Pérez Leiva, en mi calidad de profesor(a) guía/director(a) del trabajo académico mencionado anteriormente **DEJO CONSTANCIA** que:

- He revisado esta versión del documento y corresponde a la versión final aprobada del trabajo.
- El trabajo cumple con los requisitos académicos y de formato establecidos por la institución.

3.- EVALUACIÓN DE CONFIDENCIALIDAD POR PROPIEDAD INDUSTRIAL (marcar una opción)

☐ El trabajo **NO** contiene información que amerite confidencialidad y puede ser publicado de inmediato en repositorio con acceso abierto.
☐ El trabajo **CONTIENE** información con potenciales implicancias de propiedad industrial o intelectual y requiere un periodo de confidencialidad (embargo) por (marcar una opción):

☐ 6 meses ☐ 12 meses ☐ 2 años ☐ 3 años ☐ 5 años ☐ 10 años

Fundamentación de la necesidad de confidencialidad (obligatorio si se solicita embargo):

____

____

____

4.- FIRMAS

Profesor(a) guía o director(a) de memoria o tesis:

Fecha: ____________________ Firma: ____________________

Estudiante o Candidato(a):

Fecha: ____________________ Firma: ____________________

Este formulario debe ser insertado como página 2 de la memoria o tesis, completado y firmado por estudiante y profesor(a) antes de la entrega en portal PRISMA de Biblioteca USM.

This is the dedicatory page.

Acknowledgments

This is the abstract

Abstract

The global maritime industry is increasingly prioritizing sustainability, promoting the adoption of various hybrid and fully electric propulsion systems for different vessel types. This paper presents a design and sizing methodology of an energy storage system for a hybrid tugboat, customized to meet the operational demands while minimizing environmental impact. The proposed framework integrates analysis of load profile, energy storage sizing, battery technologies selection, and propulsion system optimization to achieve an optimal balance between performance, and emissions reduction. The results demonstrate that a well-designed energy storage system for a hybrid tugboat can achieve reductions in fuel consumption and greenhouse gas emissions without compromising the availability and robustness required for towing and transit operations.

Resumen

La industria marítima mundial está dando cada vez más prioridad a la sostenibilidad, promoviendo la adopción de diversos sistemas de propulsión híbridos y totalmente eléctricos para diferentes tipos de buques. En este artículo se presenta una metodología de diseño y dimensionamiento de un sistema de almacenamiento de energía para un remolcador híbrido, personalizado para satisfacer las exigencias operativas y minimizar el impacto medioambiental. El marco propuesto integra el análisis del perfil de carga, el dimensionamiento del almacenamiento de energía, la selección de tecnologías de baterías y la optimización del sistema de propulsión para lograr un equilibrio óptimo entre el rendimiento y la reducción de emisiones. Los resultados demuestran que un sistema de almacenamiento de energía bien diseñado para un remolcador híbrido puede lograr reducciones en el consumo de combustible y las emisiones de gases de efecto invernadero sin comprometer la disponibilidad y la robustez necesarias para las operaciones de remolque y tránsito.

Contents

List of Figures

List of Tables

Chapter 1

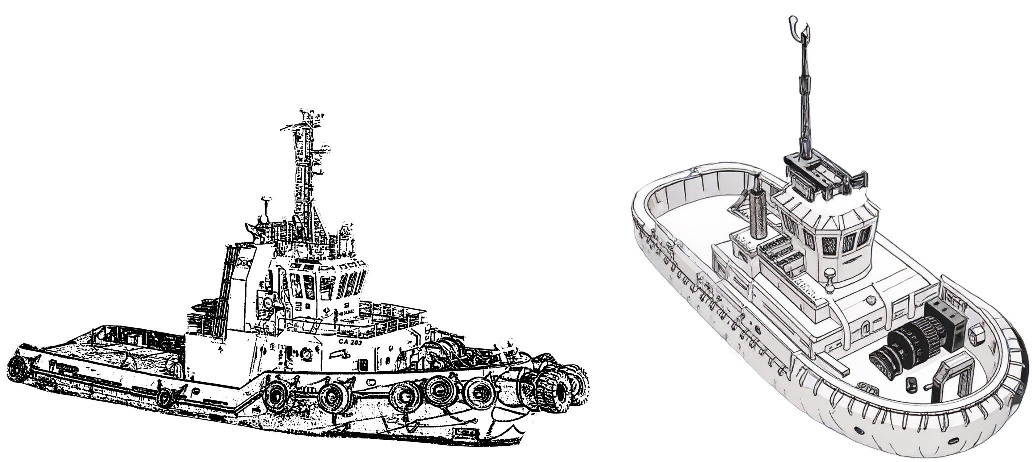
Tugboats: Description and Operation

A tug, or more commonly a tugboat, is an assistance boat which helps in the mooring or berthing operation of a ship by either towing or pushing a vessel towards the port [10] [15].

Tugboats are also essential for non-self-propelled barges, oil platforms, log rafts, etc. Due to their solid structural engineering, tugs are small but relatively powerful machines.

This chapter presents the technical and operational characteristics of tugboats, with the aim of illustrating a real service profile of a tugboat in Chile and identifying its different technical and operational aspects relevant to hybridization.

1.1 Overview of tugboats



(a) Harbour tug representation. (b) Escort Tug representation.

Figure 1.1: Tugs representations.

A tugboat is technically defined as a type of vessel specifically designed to assist other ships in moving by providing a controlled amount of force. They are essential vessels in ports, as is shown in fig. 1.1, where they assist in the movement of floating objects and in emergency tasks [29].

In maritime law, maritime towing is defined as the operation whereby a vessel (the tugboat) transports another vessel or floating device that lacks self-propulsion or, despite having it, is unable to navigate under its own power, by pulling it through the sea. This functional duality—which encompasses both high-precision maneuvering assistance in congested environments (port towing) and long-distance transport and salvage (offshore towing)—requires design engineering and system redundancy adapted to very different operational risks [31].

The importance of tugboats has grown exponentially with the phenomenon of naval gigantism [32]. Large modern ships, such as cruise ships and oil tankers, have tremendous momentum (inertia) that prevents them from stopping quickly or maneuvering effectively on their own in narrow channels or docks [30].

Therefore, tugboats are essential to counteract these inertial forces. Support is necessary for entry, exit, and general maneuvering, including assisting ships in emergency situations [31]. The efficiency of these types of vessels is based on fuel consumption in relation to load capacities, always taking into account both the propulsion system and the energy conversion system that determines it. The efficiency of conventional propulsion systems varies between 38% and 41% for a power range between 100 kW and 2,500 kW. For vessels with higher power ratings, efficiency increases to 48% [16] [17].

1.1.1 Types of tugboats

The classification of tugboats is primarily based on their operational domain and assigned primary mission. The usage and functions of tugs vary from port to port, as different ports have different requirements and intakes.

a) Harbour Tugs

Figure 1.2: A harbour tug in the port of Algeciras [34].

These vessels are specialized in assisting large ships entering and leaving port limits, as can be seen in fig. 1.2. Their design prioritizes agility and the ability to apply force in any direction with maximum efficiency [29].

- **Design:** Typical dimensions vary in length between 20 and 30 meters, with drafts of 3.0 to 4.5 meters. Their empty displacement speed ranges from 5 to 13 knots (9.3 - 24 km/h). The power of these tugboats can vary considerably, generally between 400 - 3,000 hp (300 and 2200 kW) [33].

b) Offshore/SAR Tugs

Figure 1.3: SAR BALDER, an offshore Tug built in 2006 [36].

An offshore or *search and rescue* (SAR) Tug, is designed to operate in the open sea, these tugboats are characterized by their robustness, greater autonomy, and ability to withstand severe ocean conditions. An example of this vessel is shown in fig. 1.3.

- **Design:** Typical dimensions vary in length between 25 and 70 meters, with drafts of 4.5 to 6 meters. Their empty displacement speed ranges from 5 to 13 knots (9.3 - 24 km/h). The power of these tugboats can vary considerably, generally between 1500 - 6,000 hp (1120 - 4470 kW) [33].

c) Escort Tugs

Figure 1.4: Robert Allan Rastar 3200 Escort Tug [35].

These tugboats perform a critical safety mission. Their main objective is to provide preventive escort to ships carrying hazardous materials (such as LNG or crude oil) in high-risk areas, standing ready to counteract any possible loss of control of the assisted vessel.

- **Design:** Typical dimensions vary in length between 40 and 80 meters, with drafts of 5 to 6 meters. Their empty displacement speed ranges from 15 to 16 knots (9.3 - 24 km/h). The power of these tugboats can vary considerably, generally between 6,000 -20,000 hp (4470 - 14,900 kW) [33].

d) Tugs Comparation

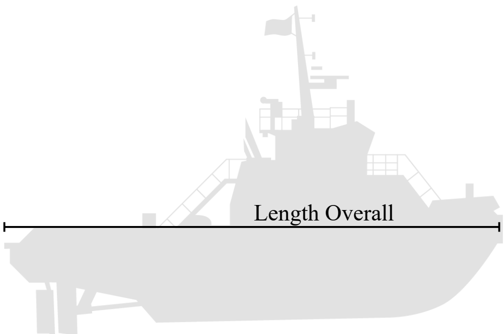
Below, in the table 1.1, there is a summary of key operational and dimensional characteristics:

Table 1.1: Operational Classification and Dimensional Parameters of the Tugboat

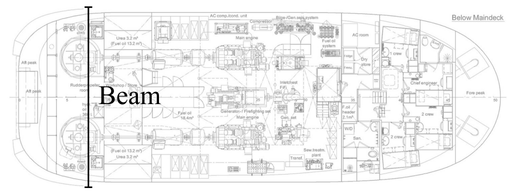
Type of Tugboat	**Operation**	**Design objective**	**Typical BP**
Harbour Tug	Protected maritime and port areas such as bays or estuaries.	Maneuverability, Reduced Draft	40-80
Offshore/SAR Tugs	Open Ocean	Autonomy, Stability, SAR Power	60-200+
Escort Tugs	High-Risk Areas (LNG, Crude Oil)	Rollover Control, Side Pull Capacity	60-120

1.2 Tugboat Sizing Characteristics

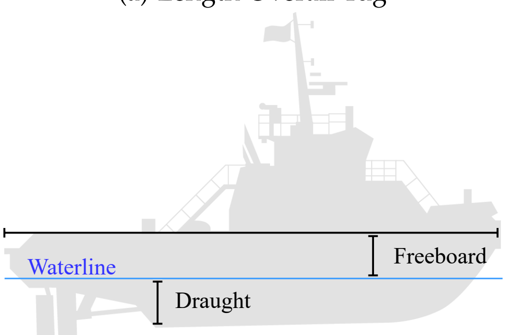
1.2.1 Dimensions



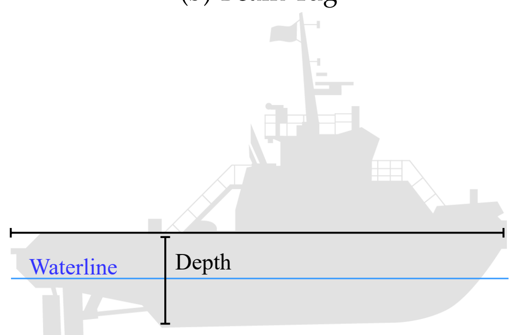
(a) Length Overall Tug



(b) Beam Tug



(c) Draught Tug



(d) Depth Tug

Figure 1.5: Tugboat Dimensions

a) Length Overall

The overall length is measured from the bow to the stern of the vessel in a straight line, excluding bowsprits, rudders, outboard motors and their supports, cranks and other accessories, fixed elements, and extensions [43], as is shown in figure 1.5a.

b) Beam

A tugboat's beam is the distance between the two widest points of the hull [44], as is shown in the figure 1.5b. It is usually measured at the waterline, although other specifications such as maximum beam or deck beam may be used. Beam is a critical dimension in naval architecture, affecting the stability, maneuverability, and overall performance of the vessel [45].

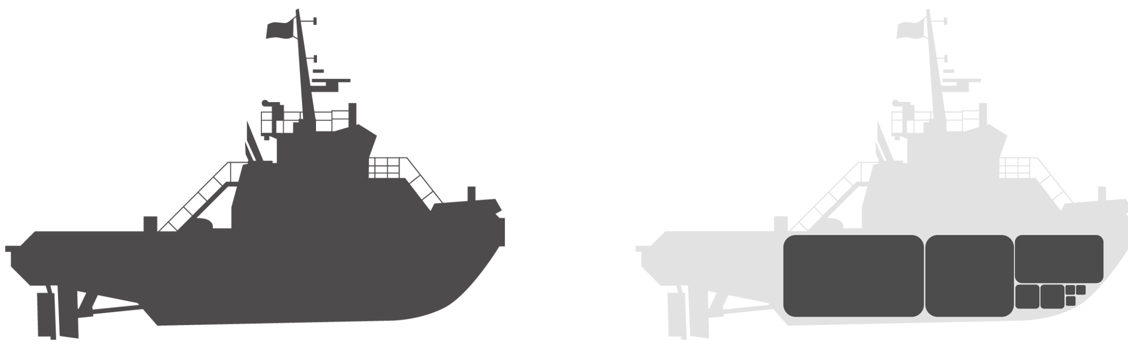
c) Draught

Draught or Draft is the vertical distance from the moulded base line amidships to the actual waterline [11], as is shown in the figure 1.5c.

d) Depth

Depth or molded depth of a vessel is measured at the center of the length, from the top of the keel to the highest continuous deck on the side [11], as is shown in the figure 1.5d.

1.2.2 Volumen and Weight



(a) Gross Tonnage Tug (b) Net Tonnage Tug

Figure 1.6: Tug Volume.

a) Gross Tonnage (GT)

Gross Tonnage (GT) measures the "total inside space" of the ship, including all areas like cargo holds, engine rooms, and crew spaces. It's about "volume", not weight [11] [46], as is shown in figure 1.6a.

- Under vessel measurement rules of various nations, the Panama Canal and the Suez Canal gross tonnage is a measure of the internal volume of space within a vessel in which 1 ton is equivalent to 2.83[m^3] or 100[ft^3].
- Under the International Convention on Tonnage Measurement of Ships, 1969, (ICTM), a standardized numerical value is a logarithmic function of spaces within a vessel. There is no definition of a ton under ICTM because the value per unit of volume is greater on a vessel of large volume than on a vessel of small volume. Gross tonnage according to the national and canal rules generally includes spaces bounded by the under surface of the uppermost complete deck, the side frames, and the floor frames or the inner bottom if it rests on the floors or if the double water is for water ballast, plus closed-in space in deck structures are available for cargo or stores or for the berthing or accommodation of passengers or crew. Rules vary greatly as to exclusion or inclusion of various spaces. Gross tonnage according to ICTM is GT = K1 V in which V is the total moulded volume of all enclosed spaces of the ship in cub m and K1 is 0.2 + 0.02 log10 V.

b) Net Tonnage (NT)

Net Tonnage (NT) is focuses on the space available for cargo [11] [46], as is shown in figure 1.6b.

- Net tonnage according to the national and canal rules is derived from gross tonnage by deducting an allowance for the propelling machinery space and certain other spaces.

- Net tonnage according to ICTM is a logarithmic function of the volume of cargo space, the draft-to-depth ratio, the number of passengers to be accommodated, and the gross tonnage.

c) Deadweight (DWT)

DWT refers to weight and indicates how much the ship can safely carry. It is expressed in long tons of 2,240 pounds (1,016.0469088 kilograms) [47].

d) Displacement Tonnage

Displacement tonnage is used to define the size of naval ships. It refers to the weight of the volume of water displaced by a vessel in normal seagoing condition [47].

1.3 Technical Characteristics of the Propulsion System

1.3.1 Bollard Pull - BP

Figure 1.7: The bollard pull test measures a boat's maximum towing capacity by securing it to a fixed point and measuring the force generated at full thrust while maintaining zero forward speed [30].

Bollard pull (BP) is the fundamental metric for quantifying a tugboat's pulling power.

BP measures the maximum force a vessel can exert while stationary in the water. It is the maritime equivalent of horsepower or torque in land vehicles. The measurement requires securing the vessel to a fixed bollard at the dock and using specialized load cells to calculate the tension generated on the line with the engines running at full power, as is shown in the figure 1.7.

BP specifications are expressed in units such as kilonewtons (kN), short tons (stf), or ton-force (tf) and are critical for demonstrating the vessel's pulling capacity under load. All commercial vessels intended for towing operations must obtain fixed point pulling certification from international classification societies such as the American Bureau of Shipping (ABS) or Bureau Veritas (BV) [32].

The tugboat's design is specifically geared toward maximizing pulling force in relation to its size. A common and effective way to increase pulling force is by adding a nozzle (duct) that surrounds the propeller, which improves propulsive efficiency [32].

1.3.2 Propulsion Engineering for Maneuvering

The maneuverability of a tugboat, which directly influences its operational efficiency and safety, is intrinsically linked to the design of its propulsion system.

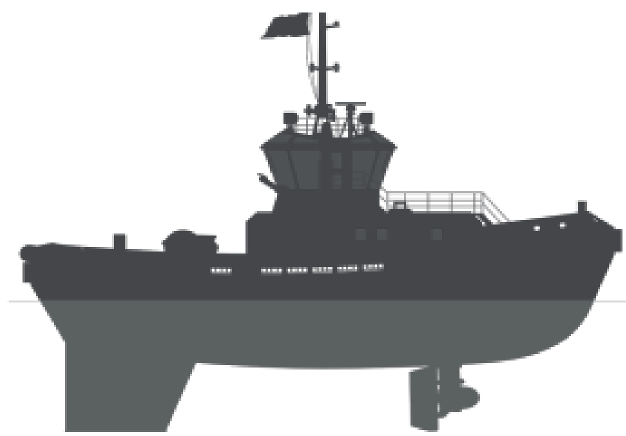
a) Conventional Propulsion

Figure 1.8: Conventional propulsion: A tug fitted with fixed propellers, single or twin screw (left or right handed) and single rudders with fixed nozzles [48].

It uses a standard diesel engine connected to a propeller, as is shown in the figure 1.8. It is a proven and simple system, with good efficiency in linear towing tasks within ports and coastal areas. However, it offers limited lateral maneuverability compared to more advanced systems [41].

b) Azimuthal Stern Drive (ASD)

(a) Azimuth Stern Drive (ASD): A tug fitted with two azimuth thrusters in nozzles at the stern, capable to direct all its thrust in any direction [48].



(b) ASD Tractor: A tug with azimuth thrusters located generally forward off the midship and a rudder-shaped fin at aft side. [48].

Figure 1.9: ASD Tugs representations.

These are rotary thrusters, where a motor is capable of directing all its thrust in any direction [41]. They all have two propellers, and since each one can be oriented independently, maneuverability is greatly increased, allowing them to turn on themselves in a single length [42], as is shown in the figure 1.9a. The only difference with conventional tugboats are the modifications to the stern and structure to house the propellers and withstand the forces exerted by the nozzle rudder, which are different from those exerted by a conventional rudder and propeller [42].

The in-line propulsion propeller (Z-Drive) is a system similar to the azimuth propulsion system, but with the difference that the propeller is oriented in only one direction (usually straight ahead), although it still provides great maneuverability in towing operations. In terms of power transmission, these tugs are more efficient [41].

An ASD Tractor Tug generally has two fixed pitch azimuth propellers in the center of the boat or slightly forward of the center of the boat, as is shown in the figure 1.9b.

The propulsion units are located well forward of the tow point, close to the pivot point, which generates a large turning thrust. This can result in poor steering performance, which is remedied by installing a large center rudder-shaped fin at aft side [49].

c) Voith-Schneider Propulsion (VSP)

Figure 1.10: Voith-Schneider Propeller (VSP) propulsion: A tug with VSP generally located forward off the midship and a rudder-shaped fin at aft side [48].

This system, as is shown in the figure 1.10, offers omnidirectional thrust and total steering control, resulting in formidable positioning accuracy. These harbor tugs have been equipped with *VoithSchneider* propellers, a unique type of propulsion system consisting of oscillating blades that allow them to have total, targeted control over the vessel without having to adjust the orientation of the main engine or propellers [41].

This propulsion system allows for highly precise operation in any direction due to its agility, making it ideal for rescue operations as well as in ports crowded with high-density maritime traffic [41].

d) Propulsion Comparation

The transition from a conventional system to an azimuth system, and even more so to a Voith-Schneider system, involves an increase in engineering complexity and acquisition cost, but this is fully justified by the need to operate with high precision and safety in confined spaces or under adverse weather conditions. The design of these advanced systems is crucial for safety, as they allow the tugboat to apply the required force (BP) without compromising lateral stability and minimizing the risk of capsizing.

Table 1.2 compares the different types of propulsion discussed in this document.

Table 1.2: Comparison of Propulsion Systems in Tugboats and Performance.

Propulsion System	Advantages	Maneuverability	Typical Application
Conventional	Low Cost, Proven Design	Limited (Good in a straight line)	Single Coastal/Offshore Trailer
ASD	High BP, Axis Rotation, Port Agility	Good	Port Towing and Escort
VSP	Omnidirectional Thrust, Extreme Stability	Very Good	Congested Ports

1.3.3 Main Diesel Engine

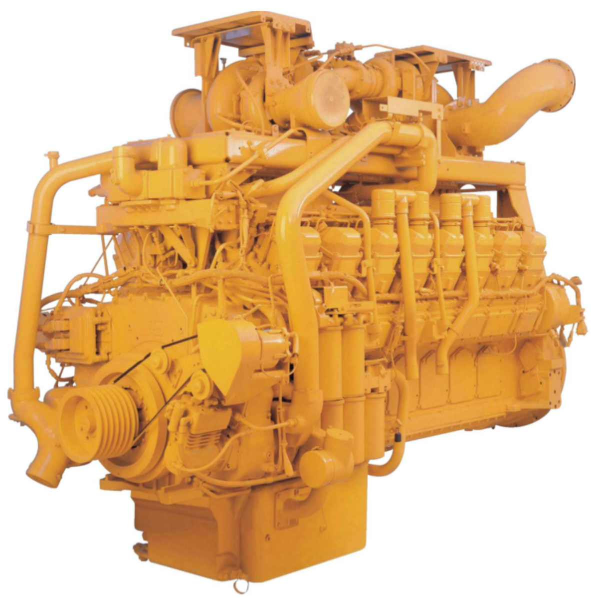


Figure 1.11: Main diesel engine for a tugboat propulsion system.

Diesel engines for propulsion, as is shown in figure 1.11, are designed to increase efficiency and keep costs to a minimum, whether in port or at a terminal. They are characterized by high power density, acceleration, and static traction.

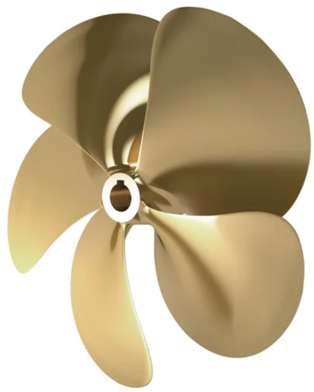
In general, the output power is available with variable load for an unlimited time, with an overload capacity of 10% of the rated power, for use in emergencies for a maximum period of time (1 out of every 12 hours, for example), considering that overload operation cannot exceed a certain number of hours per year.

Regard with the main diesel engines for propulsion system, the brake horsepower (BHP) is the power measured directly at the engine of a boat before any power loss through the transmission system or propeller shaft [50]. It influences maneuverability when docking and handling in adverse conditions.

BHP provides a realistic figure for useful power, as any kinetic application, such as a boat, will suffer a certain degree of resistance from the sea, which will reduce the actual power. Therefore, BHP can be thought of as the gross power of the engine, while the actual useful power will be somewhat lower due to mechanical losses in the transmission.

1.3.4 Conventional Propellers

a) Fixed Pitch Propellers



(a) Fixed pitch propeller.

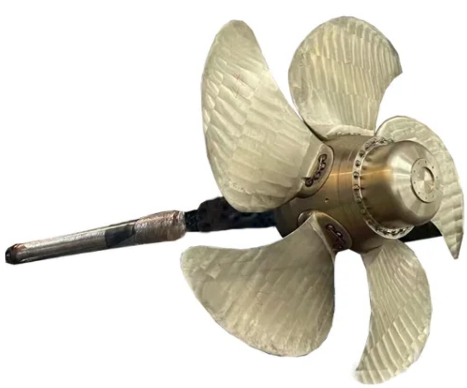
(b) Fixed pitch propeller with nozzle.

Figure 1.12: Fixed pitch propellers configuration.

These types of propellers, as is shown in the figure 1.12a, maintain their configuration unchanged, meaning that their blades do not move on their axis [33].

Also, there are fixed pitch propellers with improved propulsion efficiency thanks to the nozzle that aids vector thrust [33], as is shown in the figure 1.12b.

b) Variable Pitch Propellers



(a) Variable pitch propeller.

(b) Variable pitch propeller with nozzle.

Figure 1.13: Variable (controllable) pitch propellers configuration.

Unlike fixed-pitch propellers, in a variable pitch propeller (also known like controllable pitch proppeller) each of the blades can rotate on its own axis [33]. An example can be seen in the figure 1.13a, where each blade is attached to the central shaft by means of an anchor that allows it to rotate.

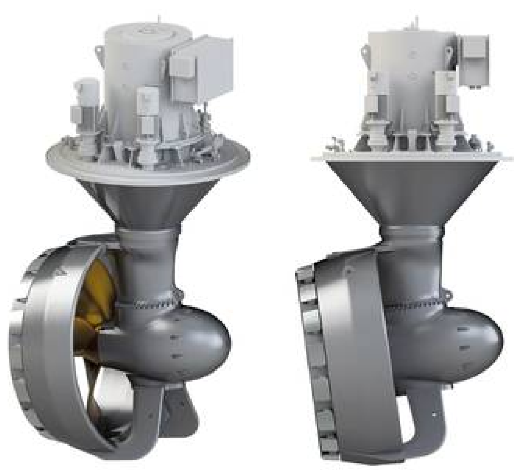
In some cases, the propulsion efficiency can be improved thanks to the nozzle that aids vector thrust [33], as is shown in the figure 1.13b.

e) Propellers Comparation

Variable pitch propellers are more efficient than fixed pitch propellers because the adjustment of the blades allows for maximum power or variable speed, which is not the case with fixed pitch propellers that are designed for specific ordinary operating conditions.

It should be noted that variable pitch propellers do not provide as much thrust for reverse navigation as fixed pitch propellers do, which limits the tasks that tugboats can perform.

1.3.5 Special Propellers



(a) Schottel system.

(b) VSP propeller.

Figure 1.14: Special Propellers.

a) Schottel system

In this system, the propeller positioned at a 90º angle to the vertical, fixed to the shaft is a nozzle inside which the propeller rotates, as is shown in the figure 1.14a. The entire assembly can rotate 360º on the vertical axis, providing excellent maneuverability and allowing movement in any direction.

b) Voith-Schneider System (Cycloidal Propeller)

This propeller system consists of a rotor that rotates on a vertical shaft fixed to the hull approximately at its pivot point, as is shown in the figure 1.14b. It is equipped with several blades that pivot on vertical axes controlled from the bridge by a device called a steering control. This device is responsible for directing the thrust in the opposite direction to the desired thrust. The mechanism is synchronized to center the thrust at a single point.

1.4 Naval Maneuvers and Assistance Techniques

Towing operations are governed by strict procedures that seek to maximize safety in dynamic and often restricted environments.

1.4.1 Docking and Undocking Procedures

The assistance of tugboats is crucial during docking and undocking maneuvers. The undocking maneuver can be as laborious as the docking maneuver and must be planned in advance. The objective is to leave the assisted vessel clear of the dock, in a safe position that allows the rudder and propeller to be effective so that the vessel can gain control and momentum.

a) Operational Coordination

The maneuver is initiated by request via VHF radio. The bow and stern operating personnel receive instructions to unmoor the lines.

b) External Agent Management

It is essential to consider the effects of winds and currents. Strong winds from outside can add to the effects of the engine in reverse during undocking, creating a dangerous situation that can violently push the stern or bow toward the dock. The ability of the highly maneuverable tugboat to apply instantaneous lateral force reduces the time the assisted vessel is exposed to the combined and unfavorable action of wind and propulsion, mitigating the risk of collision. If the wind action is severe, tugboat assistance is mandatory, especially if the actions of the assisted vessel's capstan are insufficient.

1.4.2 Tugboat Performance

The assistance of tugboats is crucial during docking and undocking maneuvers. The undocking maneuver can be as laborious as the docking maneuver and must be planned in advance. The objective is to leave the assisted vessel clear of the dock, in a safe position that allows the rudder and propeller to be effective so that the vessel can gain control and momentum.

Tugboats use various configurations to apply controlled force:

a) Tugboat Working on a Beam or Over a Cable

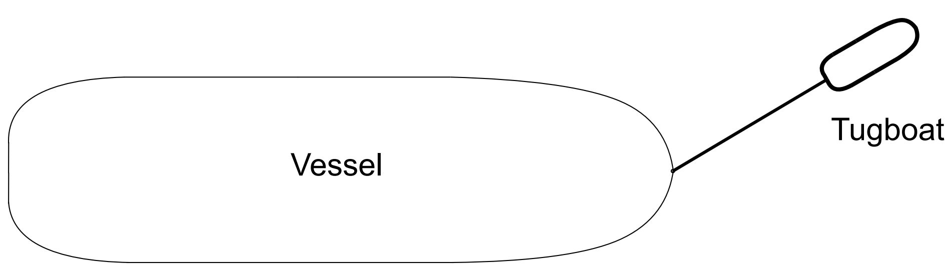


Figure 1.15: Arrow maneuver or rope maneuver [33].

In this procedure, the tugboat works separately from the vessel it is assisting, pulling it from the end of a rope, which can be attached at different points on the vessel, thus performing various functions (towing, holding, etc.), as is shwon in the figure 1.15. This procedure prevents direct contact between the two vessels and also ensures that all the power of the tugboat is exerted in the direction of the rope. The disadvantage of this procedure is that it requires more maneuvering space due to the length of the mooring line, so the system cannot be used where space is limited [33].

b) Bow-Supported Tugboat (Ram Locking)

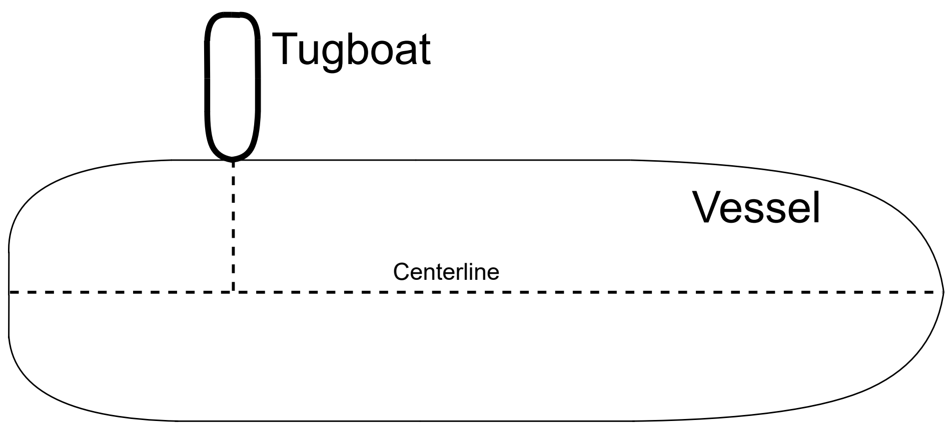


Figure 1.16: Maneuver supported by the bow [33].

In this procedure, the tugboat rests its bow on the side of the vessel it is assisting and pushes it in a direction that is essentially perpendicular to the centerline (the straight line running from stern to bow), as is shwon in the figure 1.16. It is common in this procedure for the tugboat to be secured to the ship with 1, 2, and 3 mooring lines, which prevents relative slippage between the two vessels during the maneuver [33].

c) Tugboat alongside

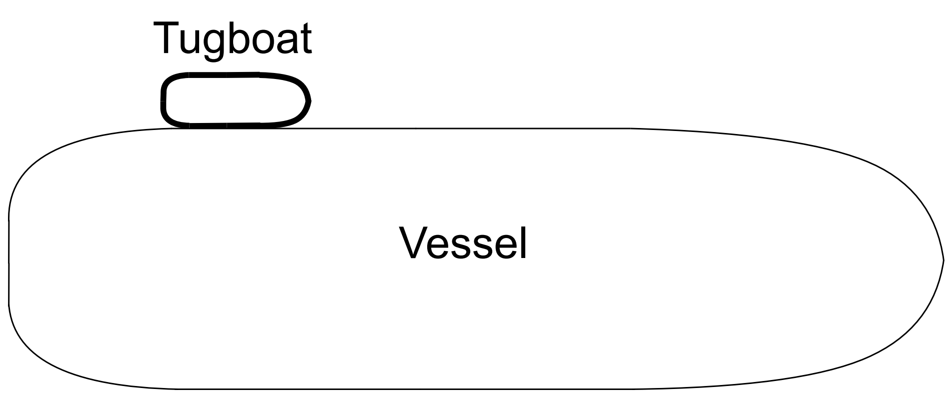


Figure 1.17: Alongside Maneuver [33].

In this procedure, the tugboat is positioned alongside the ship and roughly parallel to it, moored to the ship by several ropes that ensure the transmission of force, as is shown in the figura 1.17. This procedure is generally used to maneuver ships that do not have sufficient propulsion, in confined spaces and in very calm waters [33].

1.5 Service Maneuvers of a Tugboat

1.5.1 Operational Profile

Figure 1.18: TUG route to the Puerto Montt's pier.

Next, an analysis of a tugboat's operational load profile of a tugboat in Chile is performed, considering that it has a 10-hour work schedule, performing this task in Puerto Montt. Figure 1.18 shows the route taken by a tugboat from the shipyard to the dock in Puerto Montt. The route taken by the tugboat is divided into three different sections.

- Blue: Low-speed route, tugboat departure from shipyard.
- Yellow: Nominal propulsion speed, tugboat moves toward the dock to perform operational maneuvers.
- Red: High-power section, operational maneuver performed by a tugboat assisting a ship.

Below are some of the features associated with this tugboat:

- **Type of Tugboat**: This tugboat is considered a **harbor tugboat**, as it is focused on towing and pushing large ships that want to enter or leave Puerto Montt.

- **Dimensions**: Table 1.3 associated with the dimensions of the tugboat to be analyzed is added.

Table 1.3: Dimensions

Dimensions Tugboat	
Length overall	25 m
Beam	10.5 m
Draught	3.9 m
Depth	4.6 m

- **Volumen and Weight**: Table 1.4 contains information related to the weight and volume of the tugboat.

Table 1.4: Volume and Weight

Dimensions Tugboat	
Gross Tonnage	269 tons
Deadweight	220 tons

- **Technical Characteristics of the Propulsion System**: The tugboat has an **ASD propulsion** system, which incorporates **two Schottel propellers**. Information is shown in Table 1.5 below.

Table 1.5: Machinery

Machinery Tugboat	
Bollard Pull	65.4 tons
Main Engines	2 x Cat 3516 B
Total BHP	4894 HP (3650 kW)
Propulsion	2 x Schottel SRP 1215

- **Naval Maneuvers and Assistance Techniques**: According to the service provided by the tugboat, it is established that in the red section of Figure 1.18, the tugboat can perform operational maneuvers pulling a vessel from the end of a rope in arrow mode or **over a cable** and also perform a **alongside** maneuver, assisting ships that need to enter the port or leave the dock to set sail.

Based on the route to be analyzed and the maneuvers that the tugboat must perform at the dock, there are two operating profiles for each maneuver, as it can see in the figure 1.19 and figure 1.20.

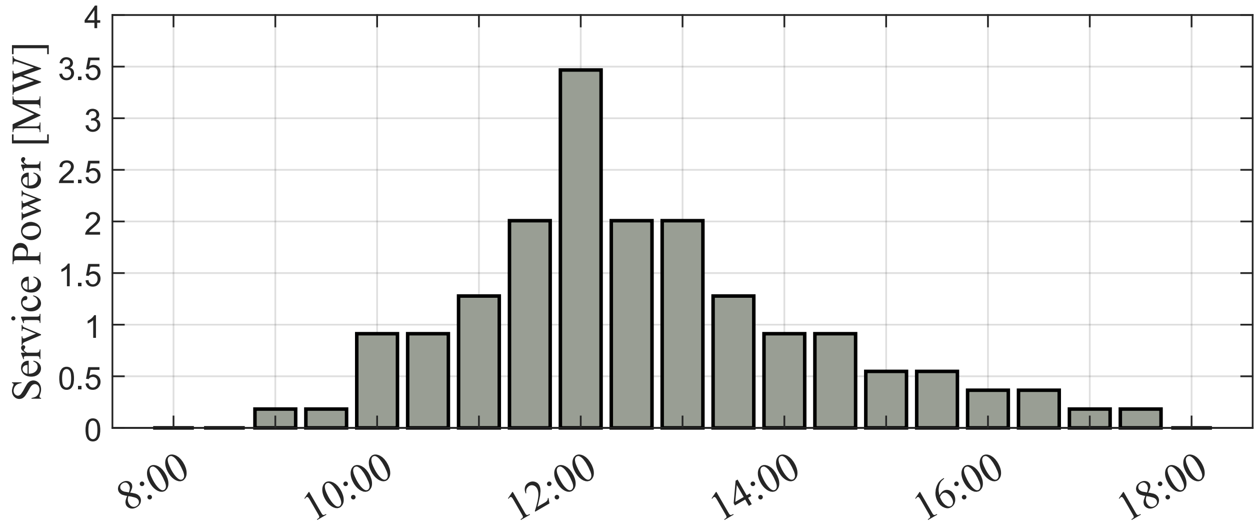


Figure 1.19: Operational profile: One maneuver.

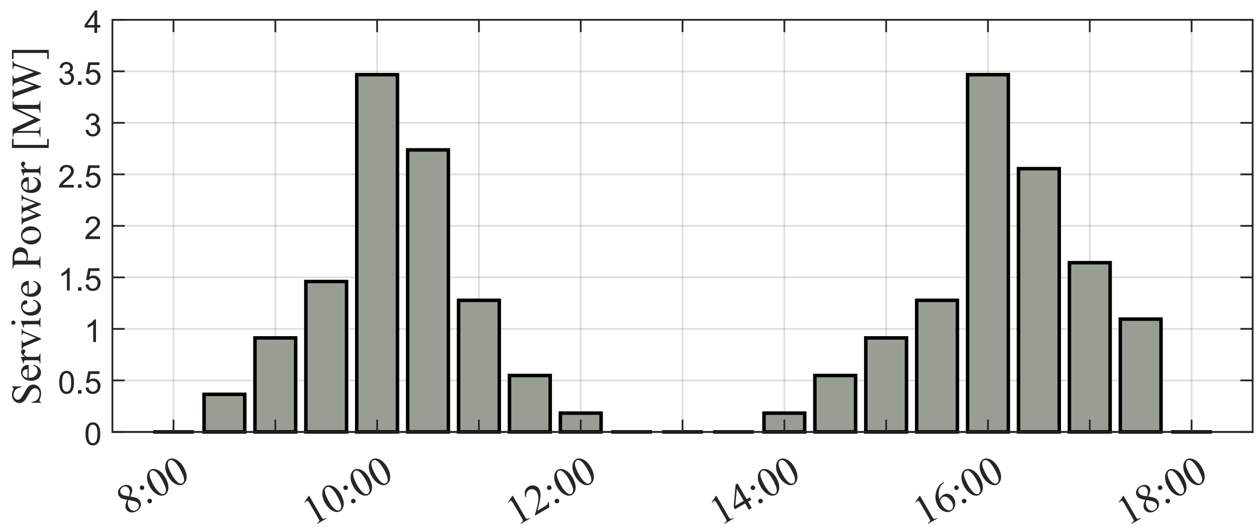


Figure 1.20: Operational profile: Two maneuvers.

Figures 1.19 and 1.20 represent the service maneuvers performed along the entire route, shown in Figure 1.18. These graphs demonstrate the power delivered by the tugboat's propulsion system.

1.5.2 Brake-Specific Fuel Consumption (BSFC)

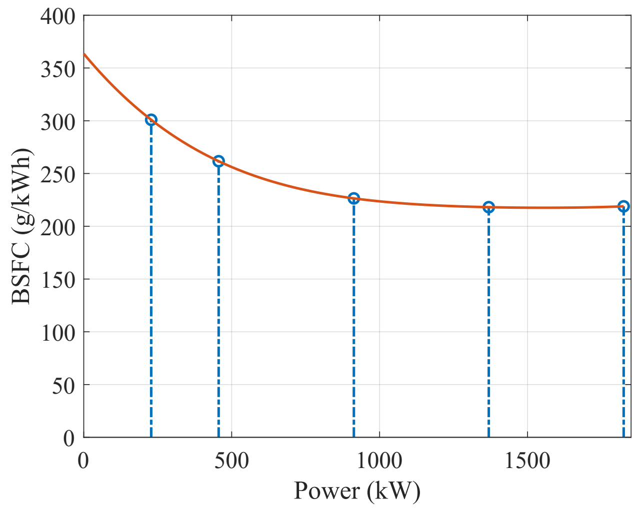


Figure 1.21: Brake-specific fuel consumption of a diesel engine model Cat 3516 B.

To analyze the operating profile, it is necessary to introduce the concept of **brake-specific fuel consumption (BSFC)**. BSFC is a measure of an engine's fuel efficiency, calculated as the amount of fuel consumed per unit of power produced over a specific time [8].

The BSFC as a function of output power is illustrated in fig. 1.21 for a typical engine used in tugboats. At higher power levels, the BSFC is lower, indicating that more power is produced per unit of fuel, resulting in greater efficiency. Maximum efficiency is achieved at the engine's rated power, where it operates as intended, maintaining an optimal balance between fuel consumption and power output. Conversely, at lower power levels, the BSFC is higher, meaning that fuel consumption per unit of power is greater, thus reducing the engine efficiency.

The relative efficiency evaluates how the conventional engine is working taking as a reference the optimal efficiency at the nominal operating point as shown in eq. (1.1) [8]. At this operating point the relative efficiency is 100% reducing its value if the engine works at lower levels of power.

$$\eta_{\mathrm{rel}} = \frac{\mathrm{BSFC}_{\mathrm{nom}}}{\mathrm{BSFC}} \tag{1.1}$$

With the information provided by the engine in Figure 1.21, it is then possible to analyze how relative efficiency varies based on the load profile. This is shown in Figures 1.22 and 1.23.

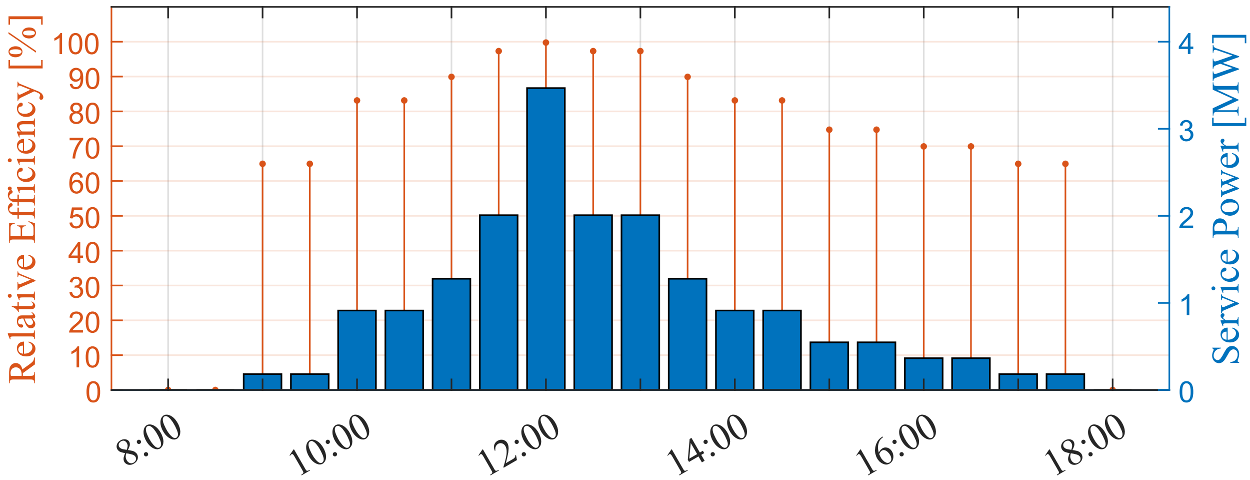


Figure 1.22: Operational profile: one maneuver.

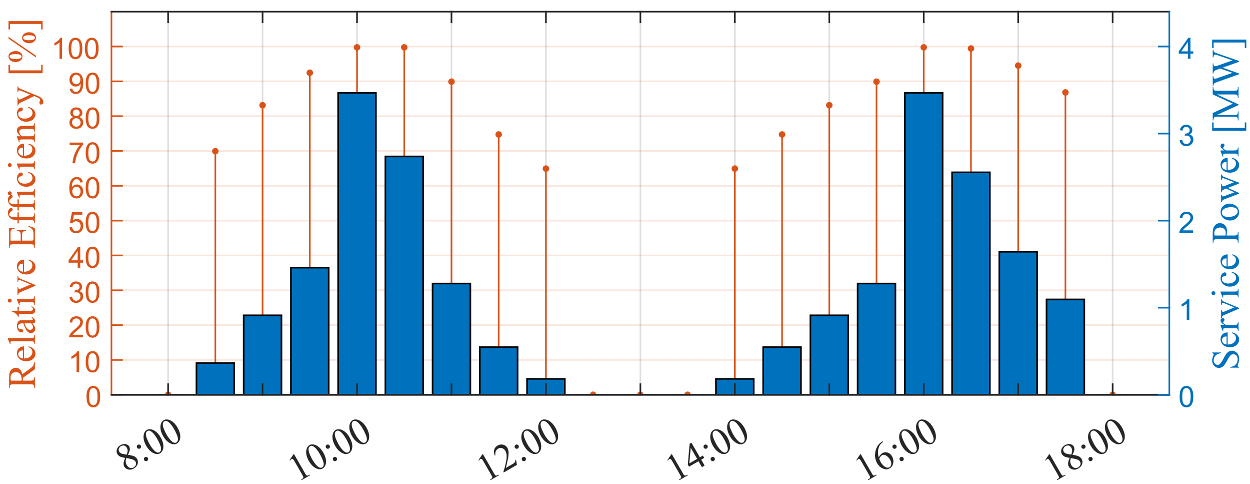


Figure 1.23: Operational profile: two maneuvers.

It can be observed in figures 1.22 y 1.23 that as the tugboat operates close to the nominal power point of its diesel engines, its relative efficiency improves. This indicates optimal fuel consumption during those periods. Conversely, at lower operating points, the engine consumes more fuel relative to the energy produced.

1.5.3 Generated Emissions and Fuel Consumption

To calculate emissions, the following formula relates brake power (P_b), BSFC, usage time (t_u), and a carbon factor (CF):

$$\text{Emissions} = \frac{\text{BSFC} \cdot P_b \cdot t_u \cdot \text{CF}}{10^{-6}} \quad [\text{t CO}_2] \tag{1.2}$$

With equation (1.2), you can find out the amount of emissions produced by the tugboat for each load profile. This can be seen in Figures 1.24 and 1.25, where fuel consumption per hour of tugboat service can also be represented, considering the density d of diesel, which in this case corresponds to 832.5 [g/L] and fuel consumption is related to emissions as follows:

$$\text{BSFC} = \frac{d \cdot \text{Fuel}}{P_b} \rightarrow \text{Emissions} = \frac{\text{Fuel} \cdot d \cdot t_u \cdot \text{CF}}{10^{-6}} \quad [\text{t CO}_2] \tag{1.3}$$

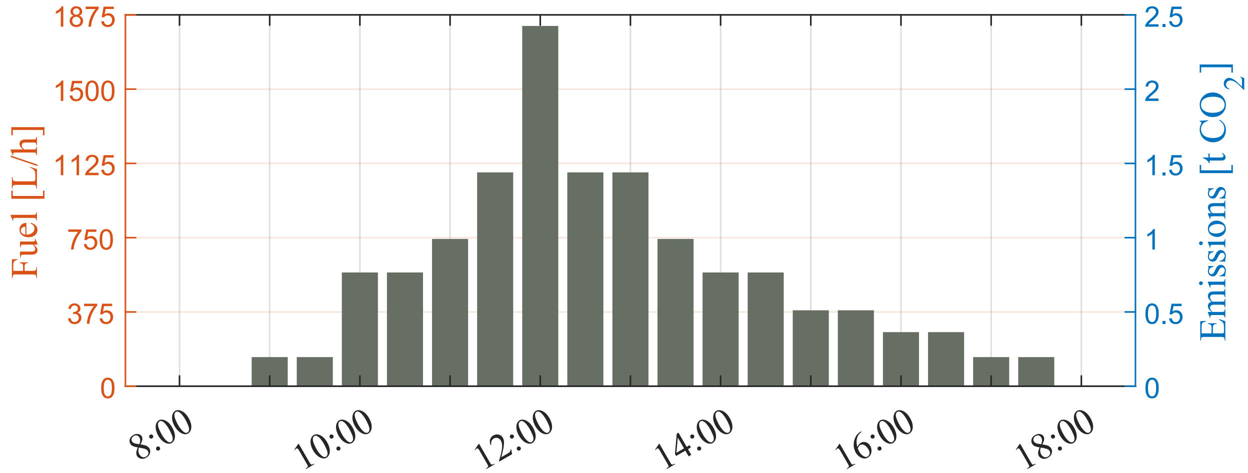


Figure 1.24: Operational profile: one maneuver, fuel consumption and emissions.

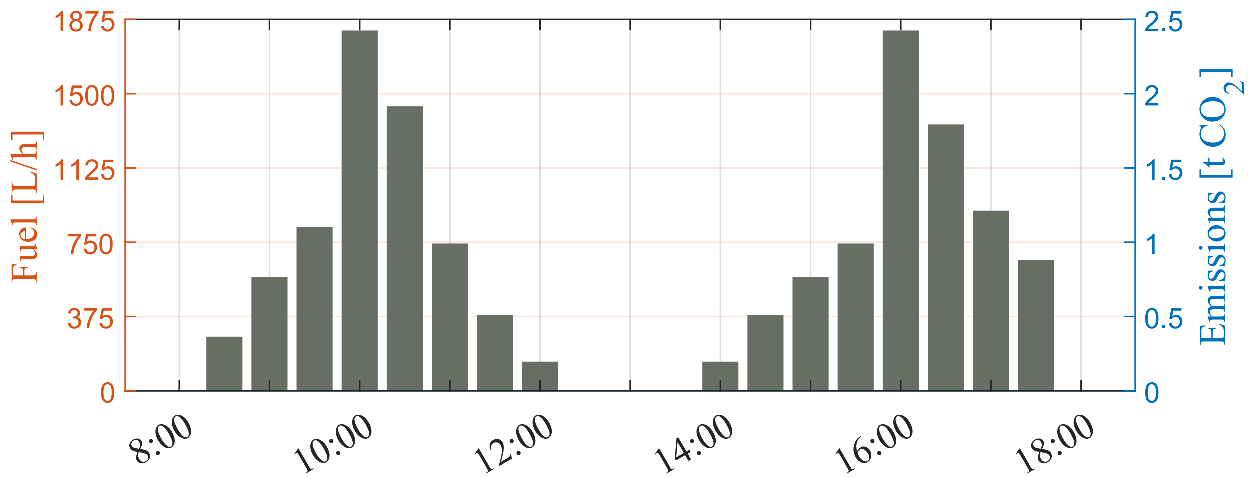


Figure 1.25: Operational profile: two maneuvers, fuel consumption and emissions.

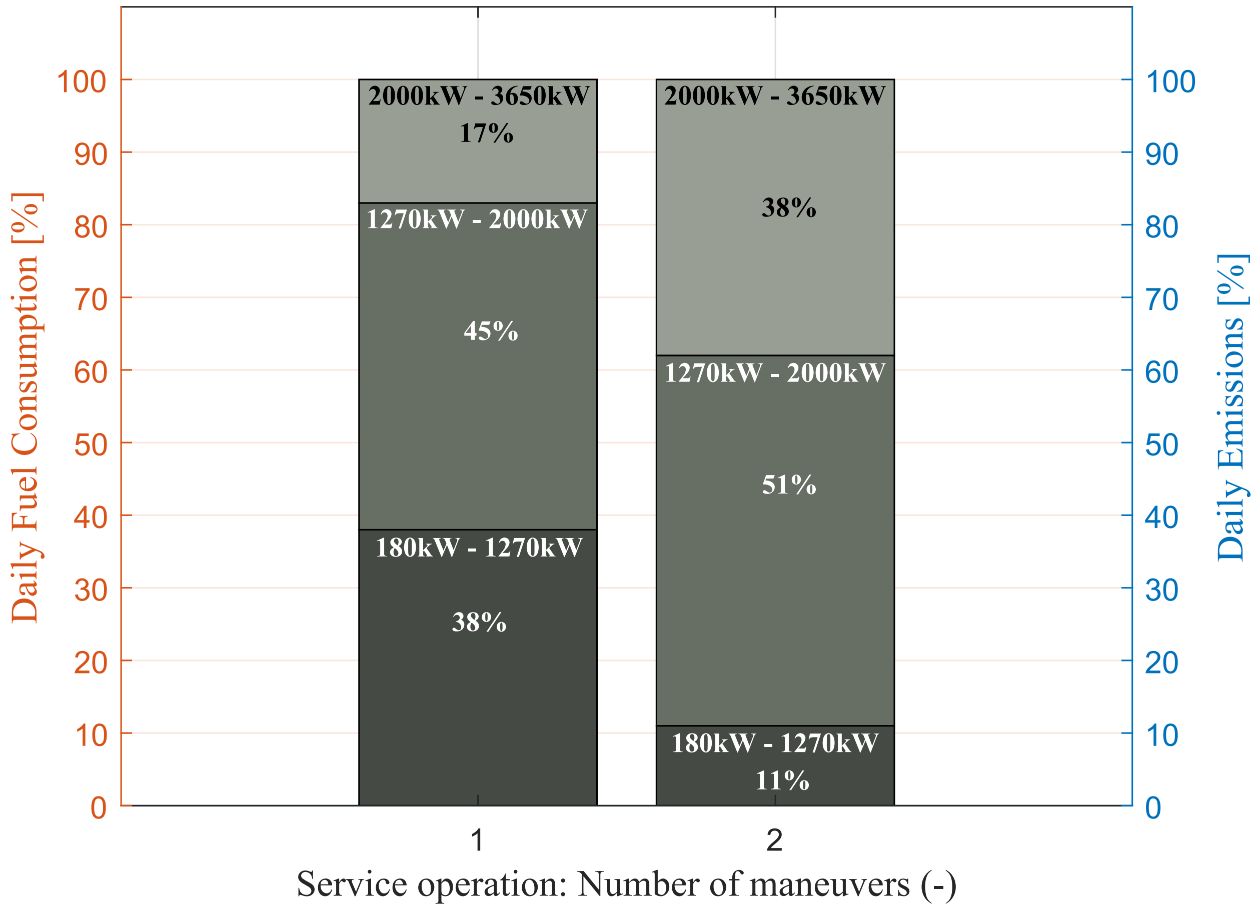


Figure 1.26: Percentage of emissions per day based on power ranges.

Fig. 1.26 shows the accumulated fuel consumption and CO_2 emissions generated for one and two maneuvers. Each service operation is divided into three segments:

- The lower segments correspond to propulsion system power levels with a relative efficiency below 90% (180kW - 1270kW).
- The middle segments correspond to relative efficiencies ranging from 90% to 98% (1270kW - 2000kW).
- The upper segments represent power levels with the highest relative efficiency, ranging from 98% to 99.8% (2000kW - 3650kW).

At low power levels, fuel consumption can account for up to 38% and 11% of the total daily fuel usage, for one and two maneuvers, respectively. Resulting in high emissions due to the diesel engine operating far from its optimal efficiency point, emphasizing the variability of the load profile of this type of vessel and its impact on fuel consumption and emissions.

1.6 Chapter Summary

This chapter established the fundamental technical and operational framework required to understand the role of tugboats in the maritime industry, serving as the baseline for the proposed hybrid propulsion design.

The tugboat is defined as an essential assistance vessel designed to aid the mooring, berthing, and maneuvering of larger ships, particularly in the context of naval gigantism. The chapter classified these vessels into three primary categories based on their operational domain:

- **Harbour Tugs**: Designed for agility and restricted drafts within ports.
- **Offshore/SAR Tugs**: Built for open ocean robustness and long-range autonomy.
- **Escort Tugs**: High-power vessels specialized in safety for hazardous cargo in high-risk zones.

A critical analysis of propulsion engineering was presented, highlighting Bollard Pull (BP) as the primary metric for quantifying a tug's static pulling force. The comparison of propulsion systems revealed distinct trade-offs:

- **Conventional Propulsion**: Offers linear efficiency but limited lateral maneuverability.
- **Azimuth Stern Drive (ASD)**: Utilizes rotatable thrusters (Z-Drive) for 360-degree maneuverability, a standard for modern port operations.
- **Voith-Schneider (VSP)**: Provides omnidirectional thrust and extreme precision for congested environments.

The chapter concluded with a case study of a tugboat operating in Puerto Montt, Chile. By analyzing the Brake-Specific Fuel Consumption (BSFC), the study identified a significant inefficiency in traditional diesel configurations:

- Diesel engines achieve optimal efficiency (low BSFC) only at high power ratings.
- The operational profile of a tugboat is highly variable; during low-load periods (e.g., transit or standby), the engines operate at low relative efficiency, leading to higher fuel consumption per kWh produced.
- Data indicated that in low power ranges (180kW – 1270kW), the relative efficiency drops significantly, yet this range can account for up to 38% of daily fuel consumption in single-maneuver scenarios

This analysis demonstrates that the intermittency of the load profile and the reliance on oversized diesel engines for low-load tasks result in excessive greenhouse gas emissions, justifying the need for the hybrid energy storage system proposed in this thesis.

Bibliography

[1] “45 Latest Greenhouse Gas & Climate Change Statistics”, TheRoundup.org [Online]: theroundup.org

[2] “How to reduce shipping emissions? Best way to cut emissions & save money in 2024”, Container xChange [Online]: container-xchange.com

[3] DNV, “Energy Transition Outlook 2025 - Maritime Forecast to 2025, A deep dive into shipping’s decarbonization journey”, 2025

[4] Völker, Thorsten. “Hybrid Propulsion Concepts on Ships.” 2013.

[5] N. Bennabi, J. F. Charpentier, H. Menana, J. Y. Billard and P. Genet, “Hybrid propulsion systems for small ships: Context and challenges,” *XXII International Conference on Electrical Machines (ICEM)*, Lausanne, Switzerland, 2016, pp. 2948-2954, doi: 10.1109/ICELMACH.2016.7732943.

[6] C. A. Reusser, H. A. Young, J. R. Perez Osses, M. A. Perez and O. J. Simmonds, “Power Electronics and Drives: Applications to Modern Ship Propulsion Systems,” in *IEEE Industrial Electronics Magazine*, vol. 14, no. 4, pp. 106-122, Dec. 2020, doi: 10.1109/MIE.2020.3002493.

[7] B. Kwasieckyj, “Hybrid Propulsion Systems: Efficiency analysis and design methodology of hybrid propulsion systems”, Master Thesis conducted at MAN Diesel & Turbo SE, Augsburg, 2013.

[8] L. Solis-Zamora, J. Perez and M. A. Perez, “Design and Sizing of an Energy Storage System for a Hybrid Tugboat,” 2025 IEEE 19th International Conference on Compatibility, Power Electronics and Power Engineering (CPE-POWERENG), Antalya, Turkiye, 2025, pp. 1-6.

[9] Lloyd Register, DNV, “ASSESSMENT OF IMO MANDATED ENERGY EFFICIENCY MEASURES FOR INTERNATIONAL SHIPPING” in Project Final Report, MEPC 63/INF.2, Annex, Page 1, 2011.

[10] SAAM, “Harbour Towage”. [Online]: saamtowage.com

[11] Wärtsilä, “Encyclopedia of Marine And Energy Technology”. [Online]: https://www.wartsila.com/encyclopedia/term/steam-turbine

[12] Maritime Education,“Marine Steam Turbines”. https://maritimeducation.com/marine-steam-turbines/

[13] Science Direct,“Gas Turbine Engines”. https://www.sciencedirect.com/topics/physics-and-astronomy/gas-turbine-engines

[14] S. Grigoryev,“Gas turbines in marine propulsion”,Bachelor’s thesis 2025, LUT University.

[15] “The Ultimate Guide to Tug Boats: Types, Functions, and Applications”, Marine Insight [Online]: marineinsight.com

[16] D. Griffiths, “Marine Medium Speed Diesel Engines”, Institute of Marine Engineers, London 1999.

[17] C. Wik, “Reducing Medium-Speed Engine Emissions”, ournal of Marine Engineering and Technology, A17, 8, 2010.

[18] P. H. Bauer and E. Pérez-Bernabeu, “The effect of the operating point choice on fuel economy in series hybrids,” 2013 World Electric Vehicle Symposium and Exhibition (EVS27), Barcelona, Spain, 2013, pp. 1-4, doi: 10.1109/EVS.2013.6914858.

[19] DNV GL AS Maritime, “Electrical Energy Storage for Ships”, EMSA European MAritime Safety Agency, 2019-0217, Rev. 04, 2020.

[20] MAN Energy Solutions, “Batteries on Board ocean-going Vessels”, 2019

[21] Chin, C.S.; Tan, Y.-J.; Kumar, M.V. “Study of Hybrid Propulsion Systems for Lower Emissions and Fuel Saving on Merchant Ship during Voyage”. J. Mar. Sci. Eng. 2022, 10, 393.

[22] Z. Fan, C. Liang, Y. Dong, G. Chang, W. Pan and S. Shao, “Control of the Diesel-Electric Hybrid Propulsion in a Practical Cruise Vessel,” 2021 IEEE International Conference on Power, Intelligent Computing and Systems (ICPICS), Shenyang, China, 2021, pp. 311-316, doi: 10.1109/ICPICS52425.2021.9524194.

[23] Kolodziejski, M.; Michalska-Pozoga, I. “Battery Energy Storage Systems in Ships’ Hybrid/Electric Propulsion Systems”. Energies 2023, 16, 1122.

[24] R. Barrera-Cardenas, O. Mo and G. Guidi, “Optimal Sizing of Battery Energy Storage Systems for Hybrid Marine Power Systems,” 2019 IEEE Electric Ship Technologies Symposium (ESTS), Washington, DC, USA, 2019, pp. 293-302.

[25] Lucà Trombetta, G.; Leonardi, S.G.; Aloisio, D.; Andaloro, L.; Sergi, F. “Lithium-Ion Batteries on Board: A Review on Their Integration for Enabling the Energy Transition in Shipping Industry”. Energies 2024, 17, 1019. https://doi.org/10.3390/en17051019

[26] DNV Maritime Forecast to 2050—Energy Transition Outlook 2022.

[27] T. Hofman, M. Naaborg and E. Sciberras, “System-Level Design Optimization of a Hybrid Tug,” 2017 IEEE Vehicle Power and Propulsion Conference (VPPC), Belfort, France, 2017, pp. 1-6, doi: 10.1109/VPPC.2017.8331009.

[28] B. A. Kumar, R. Selvaraj, K. Desingu, T. R. Chelliah and R. S. Upadhyayula, “A Coordinated Control Strategy for a Diesel-Electric Tugboat System for Improved Fuel Economy,” in IEEE Transactions on Industry Applications, vol. 56, no. 5, pp. 5439-5451, Sept.-Oct. 2020, doi: 10.1109/TIA.2020.3010779.

[29] INEAL, “Descubre las top 5 Funciones Principales del Remolcador de Puerto”. [Online]: ineal.com.mx

[30] ESC Group, “Comprender el Tirón de los Bolardos: Un Factor Clave en las Maniobras de los Barcos”. [Online]: www.escglobalgroup.com

[31] A. Bruce, A. Bermejo, “Remolque en la Mar”, Trabajo de Fin de Grado en Náutica y Transporte Marítimo, Universidad de La Laguna, sept. 2020

[32] Murimar, “Remolcadores, elemento clave en la logística marítima”, [Online]: murimar.com

[33] Ingeniero Marino, “Remolcador”,[Online]: https://ingenieromarino.com

[34] Algeciras al minuto.es,“Los remolcadores del puerto de Algeciras denuncian la excesiva carga de trabajo”,[Online]: https://www.algecirasalminuto.es

[35] Robert Allan,“Rastar Series”,[Online]: https://ral.ca

[36] Vessel Finder,“SAR BALDER”,[Online]: https://www.vesselfinder.com

[37] Prácticos de Puerto,“Remolque Portuario”,[Online]: https://www.practicosdepuerto.es

[38] Salvamento Marítimo, “Remolcadores de Salvamento”, [Online]: https://www.salvamentomaritimo.es

[39] Cotecmar, “Remolcado Offshore”, [Online]: https://www.cotecmar.com

[40] Portal Portuario, “SAAM recibe dos nuevos remolcadores para reforzar su flota en Chile, Mayo 2016”, [Online]: https://portalportuario.cl

[41] M. Jiménez Fernández, Dr. J. González Almeida, “Remolcadores: Diseño y Desempeño,” Grado en Náutica y Transporte Marítimo, Escuela Politécnica Superior de Ingeniería Sección Náutica, Máquinas y Radioelectrónica Naval, Universidad de La Laguna, Marzo de 2025.

[42] J. Farina Orihuela, A. Poleo Mora, “Maniobras con Remolcadores ” Grado en Náutica y Transporte Marítimo, Escuela Politécnica Superior de Ingeniería Sección Náutica, Máquinas y Radioelectrónica Naval, Universidad de La Laguna, Junio de 2015.

[43] wle online, “Measuring Length Overall”, [Online]: https://wleonline.com

[44] HZ-Containers, “Glosario: Beam-Ancho del Barco”, [Online]: https://hz-containers.com

[45] Discover Boating, “What Is a Boat Beam and Why Is It Important?”, [Online]: https://www.discoverboating.com

[46] Marine Rubber Fender, “Difference between Gross Tonnage, Net Tonnage, and Deadweight Tonnage”. [Online]: https://www.chinarubberfender.com

[47] Britannica, “Cars and Other Vehicles: Tonnage”, [Online]: https://www.britannica.com

[48] G. Çakıroğlu, B. Sener, A. Balin, “Applying a fuzzy-ahp for the selection of a suitable tugboat based on propulsion system type”. Brodogradnja. 69. 1-13. 10.21278/brod69401, 2018.

[49] Shiphandling for Professionals, “Types of Tugs”, [Online]: https://www.shiphandlingpro.com

[50] Yacht Buyer, “Horsepower vs Brake Horsepower (hp vs bhp): What’s the Difference?”, [Online]: https://www.yachtbuyer.com